Indigibilli Harshit

Software Engineer · Full-Stack · AI/ML · Data Science

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SUMMARY

Software engineer with experience across full-stack development, machine learning and data science. Builds backend services and web applications with Python, FastAPI, Java and React, and develops deep learning models with PyTorch and TensorFlow for computer vision and time-series problems. Experienced in turning large real-world datasets into dashboards and production features. IEEE-published author and two-time hackathon winner.

EXPERIENCE

EarthSense Labs | *Software Development Engineer (SDE)* Jul 2026 – Present

- Core team member on GeoTwin 4D, a geospatial digital twin platform for cities, infrastructure and natural environments.
- Work on its multi-tenant, multi-backend architecture (Python, FastAPI, Java) with multi-tenant access control.
- Platform built on open geospatial standards: 3D Tiles, Cloud Optimized GeoTIFF (COG), Potree point clouds and OGC services.

HydroSense Lab, IIT Delhi | *Research Intern, under Prof. Manabendra Saharia* Oct 2025 – Jun 2026

- Started as a proof of concept from a research hiring post; grew into a six-month geospatial research internship.
- Built the IPED dashboard: FastAPI backend and React (TypeScript) frontend to explore 32 years (1991–2023) of daily gridded precipitation data developed with the India Meteorological Department (IMD).
- Built real-time analysis and visualisation components for terabytes of spatial data in digital twin applications.

NIT Rourkela | *Summer Research Intern, Dept. of ECE, under Dr. Shrishail Hiremath* May 2025 – Jul 2025

- Configured software defined radio (SDR) hardware with GNU Radio for real-time signal acquisition and processing pipelines.
- Collected 70+ GB of RF data from real-world transmitters; trained deep learning models for IoT RF fingerprinting (RFFI).

The Social Ants | *AI and ML Engineer Intern* Jan 2025 – May 2025

- Built predictive analytics models for customer behaviour and integrated them into production workflows.
- Implemented AI-driven marketing automation and optimised data pipelines for real-time insights.

PROJECTS

RHPA: Roadkill Hotspots and Policy Advocacy | *Flask, JavaScript, YOLO, DenseNet, ResNet*

- Flask web app with DenseNet/ResNet classifiers and a YOLO real-time detector for roadside wildlife alerts; in IEEE Xplore.

Vantage: Productivity Analytics Dashboard | *React (TypeScript), FastAPI, Firestore, Google Calendar API*

- Live Google Calendar data: OAuth sign-in, incremental sync, recurring-event expansion, Firestore completion tracking.
- Analytics suite: adherence and focused-hours trends, category analysis, peak productivity hours, calendar heatmap.

Land Cover Classification, Delhi-NCR | *Python, ResNet18, Sentinel-2*

- Grid generation, spatial filtering, ESA WorldCover labelling and ResNet18 training on Sentinel-2, with a map layer.

Sleep Breathing Irregularity Detection | *Python, 1D CNN, Conv-LSTM, time series*

- Classifies apnea, hypopnea and normal breathing from airflow, thoracic and SpO_2 signals; leave-one-participant-out CV.

SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C, C++
Software and full-stack: FastAPI, Flask, Django, React (TypeScript), REST APIs, microservices, multi-tenant architecture
AI and ML: PyTorch, TensorFlow, computer vision (YOLO, ResNet, DenseNet), LSTM, Conv-LSTM, signal processing
Data science and analytics: data pipelines, predictive analytics, time series, geospatial data (Sentinel-2, ESA WorldCover, COG, 3D Tiles, OGC), Matplotlib, Seaborn, Power BI, Tableau
Tools and cloud: Git, GitHub Actions, Linux servers, Azure, Netlify
CS fundamentals: data structures and algorithms, OOP, DBMS, operating systems, computer networks

PUBLICATION

L. Patra, **I. Harshit**, S. K. Mohapatra, A. Singh, M. N. Mohanty, "Roadkill Hotspots and Policy Advocacy (RHPA): A Multi-Layered AI and IoT Approach for Real-Time Wildlife Preservation," *ESIC 2026*, Bhubaneswar, pp. 139–141. doi:10.1109/ESIC68176.2026.11495994

EDUCATION

Sri Sri University, Cuttack, Odisha | B.Tech, Computer Science and Engineering (Data Science) | CGPA 8.66 Sep 2022 – Apr 2026

ACHIEVEMENTS

- 1st place, IBM ICE Ideathon 2025 (RHPA)
- 1st place, WATCH Hackathon 2024, Sustainable Cities (Eco-Yatra)
- Runner-up, IBM ICE Technical Paper Presentation 2024 (data-driven drug discovery)
- Finalist, Action Plan Social Innovation Track, IIT Bombay 2024